

GAZANIA MARINE JYRWA

Indian Institute of Technology, Jodhpur, India

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RESEARCH INTERESTS

Scientific Machine Learning, with a growing interest on Fluid Dynamics as an application domain; Neural Operators and Physics-Informed Architectures for Complex Flows; Scientific Foundation Models.

RESEARCH EXPERIENCE

Indian Institute of Science, Bangalore

November 2025 – May 2026

Research Associate, Advisor: Prof. Sashikumaar Ganesan

- Contributed to AW-FNO, a gated dual-branch neural operator (Fourier + Wavelet) for turbulent flow super-resolution, outperforming standalone FNO/WNO across four PDE benchmarks. Github
- Core contributor to SciREX a modular scientific computing ML library, designing scalable architecture used in research and industrial projects.
- Building a transformer-based Scientific Foundation Model (India AI Mission) pre-trained on multi-physics PDE data with pipelines linking simulations to LLM-generated physics explanations.

Project Assistant, Advisor: Prof. Sashikumaar Ganesan

August 2025 – November 2025

- Industry Collaboration (Geometry-Aware Neural Operators): Developed Geometry-Informed Neural Operators (GINOs) predicting magnetic fields on irregular stator geometries an encoding extensible to deformable-object dynamics.

Indian Institute of Technology Jodhpur

August 2024 – May 2025

MSc Project, Advisor: Prof. Kirankumar R. Hiremath

- Conducted a comparative study of Finite Difference Methods (FDM) and Physics-Informed Neural Networks (PINNs) for solving the Helmholtz equation (elliptic PDE), evaluating accuracy, convergence, and computational efficiency through systematic numerical experiments.

WORK EXPERIENCE

ZenteiQ AiTech Innovations Pvt. Ltd.

June 2026 – July 2026

SciML Research Consultant

- Implementing Transformer-based PDE solvers, Transolver in JAX and exploring physics-attention mechanisms for learning over spatial point representations, while contributing to production-grade SciML models for industrial applications.

PUBLICATIONS

AW-FNO: Adaptive Wavelet-Fourier Neural Operator for Fluid Flow Super-Resolution

Accepted at the 13th International Conference on Computational Fluid Dynamics (ICCFD13), 2026

TECHNICAL SKILLS

- **Languages:** Python, C++
- **Libraries & Frameworks:** PyTorch, JAX, TensorFlow, NumPy, Pandas, Matplotlib, SciPy
- **Deep Learning & Computer Vision:** CNNs, Image Analysis, attention mechanisms, representation learning

- **Scientific Computing & SciML:** Neural Operators, PINNs, FDM, GPU computing, CUDA, Gmsh
- **Tools & Workflow:** Git, GitHub, Github Actions, Docker, Linux, ClearML, L^AT_EX, VSCode, Jupyter

EDUCATION

Indian Institute of Technology, Jodhpur
Master of Science (M.Sc) in Mathematics

August 2023 - May 2025
CGPA: 6.29/10

North Eastern Hills University, Shillong
Bachelor of Science (B.Sc) in Mathematics

August 2020 - July 2023
Percentage: 65.5

HIGHLIGHTED PROJECTS

Transolver: 2D Turbulence & Aerodynamic RANS Surrogates

Github

Benchmarked Transolver against FNO on 2D turbulence and unstructured-grid RANS aerodynamics

- Transolver achieved 0.40 % relative L error at 183k parameters vs FNO's 0.46 % error at 1.42M parameters
a $1.5\times$ accuracy gain with $10\times$ parameter efficiency

PINSTT: Physics-Informed Neural Spatiotemporal Tubes

Github

Reproduced paper arXiv:2512.08248 from scratch: learning time-varying safety tubes for robot navigation using PINNs with formal verification guarantees

- Implemented a three-stage pipeline: PINN training, Lipschitz-based formal verification, and a log-barrier online controller for Temporal Reach-Avoid-Stay (T-RAS) tasks.

Wavelet variant of Geometry informed Neural Operator GINO

Github

Implemented a wavelet-based extension of Geometry-Informed Neural Operators, replacing Fourier kernels with localized multi-resolution operators

- Observing stability and accuracy on irregular geometries and localized solution structures.

Helmholtz Equation Solver: PINNs vs FDM Github

Github

Implemented and compared Physics-Informed Neural Networks (PINNs) and Finite Difference Methods (FDM) for solving the Helmholtz equation

- Analyzed error, convergence, and computational efficiency against analytical solutions.

RELEVANT COURSEWORK

Mathematics: Linear Algebra*, Optimization*, Probability & Statistics*, Vector Calculus, ODEs*, PDEs*, Numerical Methods, Dynamical Systems*

Computational Science & Programming: Digital Image Analysis*, Machine Learning and Pattern Recognition*, Programming Techniques*, Data structures & Algorithms*

Online Courses: Machine Learning Specialization (Coursera), Neural Networks from Scratch (Udemy), Python for Data Science and Data Analysis (Udemy), Practical Deep Learning (Udemy)

(* denotes postgraduate-level subjects taken as major coursework)

EXTRA-CURRICULARS

- Captain, Women's Football Team, IIT Jodhpur — Runners-up, Varchas (2024)
- Active athlete with multiple medals in collegiate and school sports and athletics.